STATISTICS COURSE PROJECT | DATASET A | NHANES 2021-2023

Phase 1 Report
Data Understanding and Descriptive Statistics

Educational attainment and positive depression screening among U.S. adults aged 20 years and older

Research question

Does the prevalence of positive PHQ-9 screening differ across education levels, and does the association remain after adjustment for age, gender, and the family income-to-poverty ratio?

6,064	5,208	4,584
Adults aged 20+	Primary analytical sample	Complete-case sample

Scope: 10 common Phase 1 tasks plus 4 NHANES-specific tasks (A1-A4). The report documents descriptive findings only. Significance testing and adjusted models belong to later phases.

Data: NHANES August 2021-August 2023, DEMO_L and DPQ_L public-use files.

Analysis: Python notebook (**notebooks/phase1.ipynb**) and associated CSV/figure outputs.

1. Research framework and study design

Task 1 - Data provenance and collection

The National Health and Nutrition Examination Survey (NHANES) is conducted by the National Center for Health Statistics (NCHS), part of the U.S. Centers for Disease Control and Prevention. The analysis uses the August 2021-August 2023 cycle. Demographic information comes from DEMO_L.xpt; depression screening responses come from DPQ_L.xpt. NHANES combines interviews, questionnaires, and mobile examination center assessments using a complex probability sample.

The public-use files provide de-identified participant records; restricted information is not part of this analysis. The DPQ measures symptoms over the preceding two weeks; it is not a clinical diagnosis.

Task 2 - Unit of observation and record structure

One record represents one participant, identified by SEQN. Within the used files, SEQN is unique and the DEMO-DPQ merge uses a validated one-to-one join. The merged file contains 6,337 individuals and no repeated observations of the same participant.

Task 3 - Selection, design, and generalizability

NHANES employs a stratified, clustered, multistage probability sample, not simple random sampling. The research design is cross-sectional and observational: respondents are not randomly assigned educational levels. The intended target population is civilian, noninstitutionalized U.S. adults aged 20 years and older, subject to eligibility, nonresponse, and the analysis-specific exclusions.

Weighted percentages account for unequal sampling contributions, but survey-design variables are additionally required for valid design-based uncertainty estimates. Because complete PHQ-9 and income requirements exclude some participants, weighted results require cautious interpretation. No causal inference is made.

Task 4 - Measurement scales and roles

Variable	Scale	Analytical role
Educational attainment (DMDEDUC2_clean)	Ordinal, 5 categories	Main explanatory variable
PHQ-9 positive (score >= 10)	Binary	Primary outcome
PHQ-9 total, 0-27	Discrete quantitative	Secondary outcome
Age (RIDAGEYR)	Quantitative in years	Adjustment covariate
Recorded sex (RIAGENDR)	Nominal categorical	Adjustment covariate
Income-to-poverty ratio (INDFMPIR)	Quantitative, top coded at 5	Adjustment covariate
WTMEC2YR; SDMVSTRA; SDMVPSU	Design fields	Weight, stratum, PSU

PHQ-9 items DPQ010-DPQ090 each have ordered responses 0-3. Treating education codes as equally spaced numerical values would not be justified without additional assumptions.

2. Data quality and empirical distributions

Task 5 - Data-quality audit

The audit covers missing responses, special codes, participant uniqueness, range checks, and logical consistency. PHQ-9 item responses coded 7 (refused) and 9 (do not know) are treated as missing. The total score is calculated only when all nine items are available. Very small numerical representations of zero from XPT import are normalized to zero.

Criterion (age 20+ sample)	Count	Handling
Incomplete PHQ-9 items / total	854	Excluded from PHQ-9 total analyses
Missing or invalid education	4	Excluded from education comparison; 2 overlap with incomplete PHQ-9
Missing income-to-poverty ratio	792	Retained where income is unnecessary; 624 missing in primary sample
Duplicated SEQN	0	Unique records confirmed
Nonpositive or missing MEC weights	0	No such records in analytical sample

There were no reported invalid values in the key cleaned demographic/PHQ-9 fields checked by the audit. Missingness in unrelated retained NHANES columns is not equivalent to invalid analytical records.

Task 6 - Frequencies and empirical distributions

The primary descriptive dataset has 5,208 eligible participants with complete PHQ-9 and valid education. Unweighted frequency distributions are presented below; percentages refer to this analytical subset.

Education category	n	Share (%)	Positive n
Less than 9th grade	176	3.38	35
9th-11th grade	372	7.14	61
High school graduate/GED	1,071	20.56	165
Some college/AA degree	1,622	31.14	245
College graduate or above	1,967	37.77	154
Total	5,208	100.00	660

The distribution of screening status is 4,548 negative (87.33%) versus 660 positive (12.67%). Recorded sex: 2,363 male participants (45.37%) and 2,845 female participants (54.63%). All are unweighted sample proportions.

3. Central tendency, variability, and distributions

Task 7 - Measures of center and dispersion

Metric	Age (years)	Income-to-poverty	PHQ-9 total
Valid N	5,208	4,584	5,208
Mean	53.72	3.02	4.04
Median	58.00	3.00	2.00
Standard deviation	17.04	1.65	4.68
First quartile (Q1)	39.00	1.52	1.00
Third quartile (Q3)	68.00	5.00	6.00
IQR (Q3 - Q1)	29.00	3.48	5.00
Observed range	20-80	0-5	0-26

The PHQ-9 distribution is right-skewed: its mean (4.04) exceeds its median (2), so the median and IQR complement the mean and SD. The income-to-poverty measure is top-coded at 5 in the public data. Age includes values top-coded at 80, limiting interpretation at the upper end. These are unweighted descriptive summaries.

Task 8 - Appropriate visualizations (Figure 1)

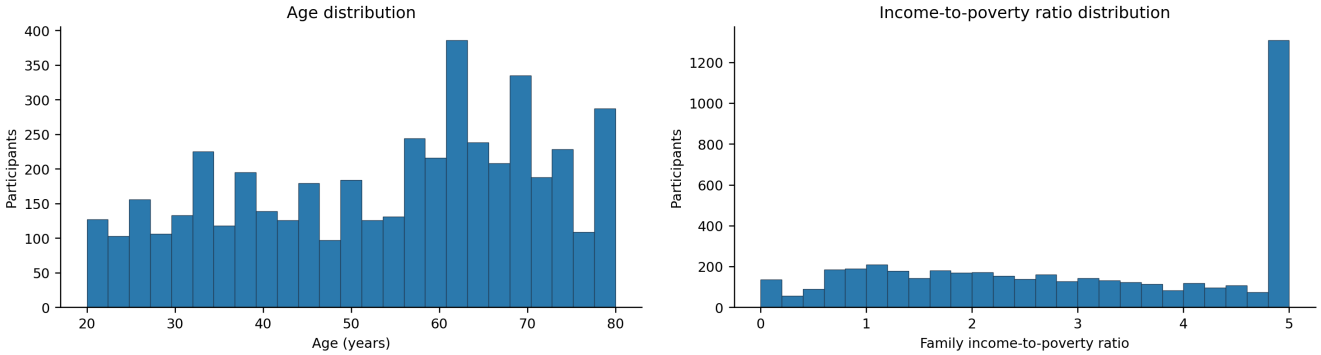


Figure 1. Age and family income-to-poverty ratio for the complete-case sample (n = 4,584); histogram counts are unweighted. The concentration at income ratio 5 reflects public-data top coding.

Histograms match the quantitative measurement scales. Axis titles identify measurement units, and bar counts begin at zero. The other figures compare the binary screening outcome by education and recorded sex.

4. PHQ-9 screening and education patterns

Task 8 - PHQ-9 distribution (Figure 2)

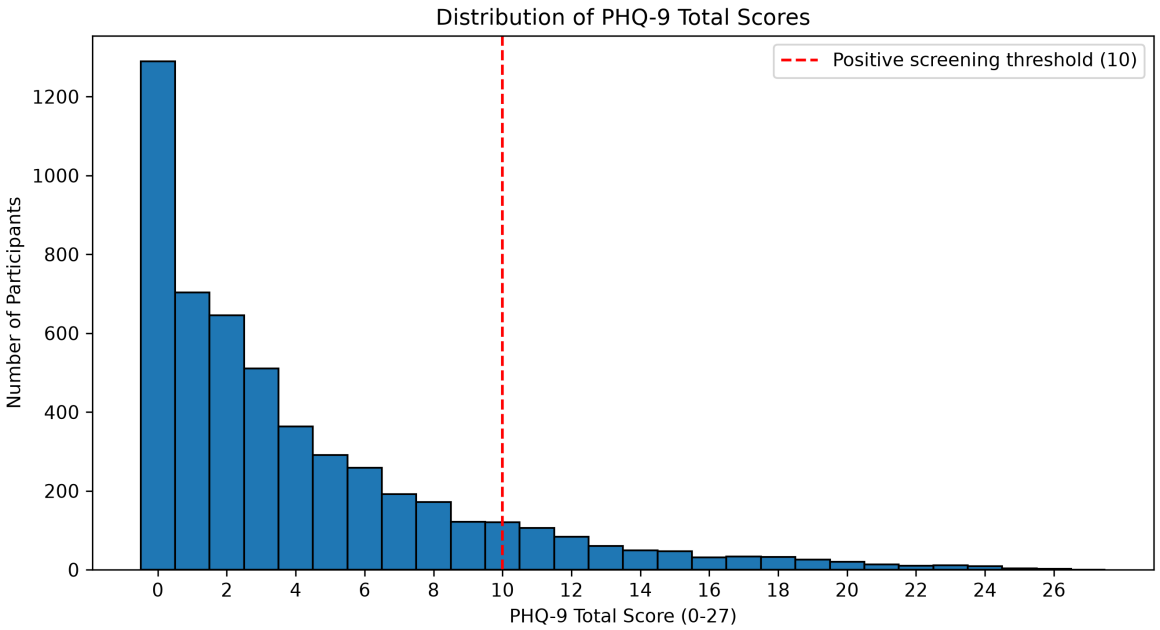


Figure 2. Distribution of PHQ-9 total scores (n = 5,208). The dashed vertical line marks the positive-screening threshold of 10; counts are unweighted.

Task 8 - Education-group comparison (Figure 3)

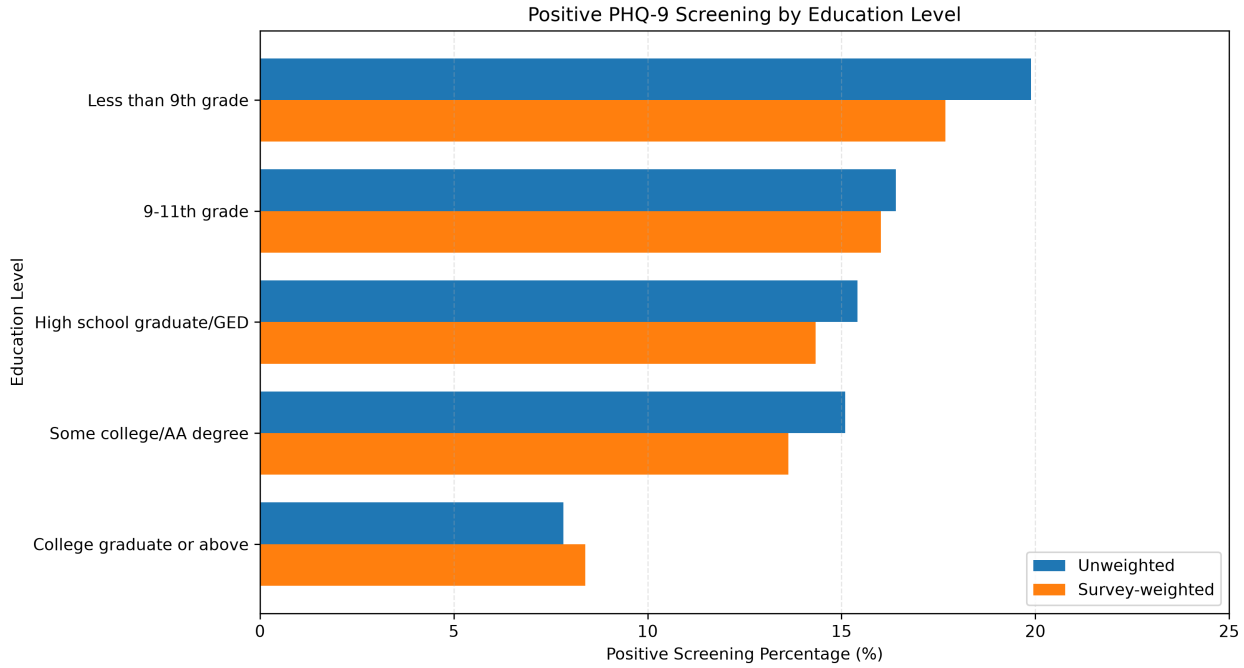


Figure 3. Percentage screening positive by educational attainment, unweighted versus MEC-weighted, across five ordered educational categories.

5. Group comparison and descriptive interpretation

Task 8 - Recorded sex comparison (Figure 4)

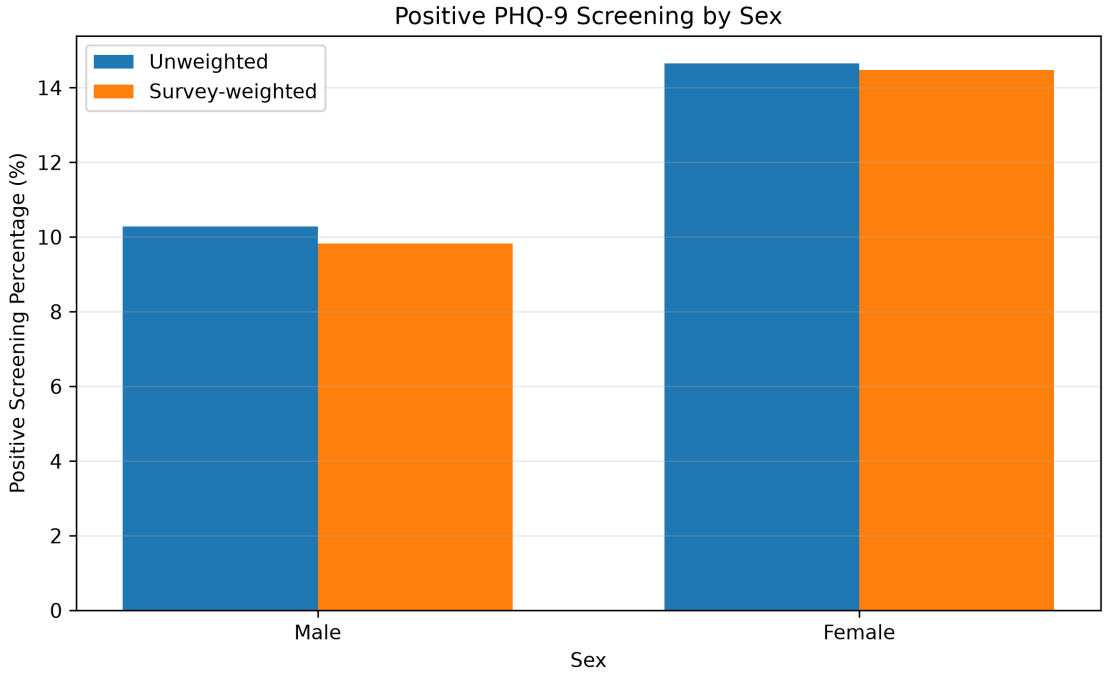


Figure 4. Positive PHQ-9 screening by recorded sex in the main analytical sample; descriptive weighted and unweighted percentages.

Task 9 - Preliminary comparison by educational attainment

Education	Positive n / N	Raw (%)	Weighted (%)
Less than 9th grade	35 / 176	19.89	17.68
9th-11th grade	61 / 372	16.40	16.02
High school/GED	165 / 1,071	15.41	14.33
Some college/AA	245 / 1,622	15.10	13.63
College graduate+	154 / 1,967	7.83	8.39

The observed prevalence generally decreases across education categories. The raw difference between the lowest and highest education groups is 12.06 percentage points; the weighted difference is 9.29 percentage points. These contrasts are descriptive, not tests of statistical significance, and may be confounded by age, recorded sex, and income.

6. NHANES-specific requirements: merge and selection

Task A1 - Source files and merge key

Both public-use files are linked by SEQN. Each input contains one record per SEQN, and the final inner merge is validated as one-to-one. A preliminary outer merge was used to assess match coverage.

Join membership	Distinct people
DEMO_L.xpt	11,933
DPQ_L.xpt	6,337
In both (inner join)	6,337
DEMO only	5,596
DPQ only	0

The 5,596 DEMO-only participants include people not eligible for the adult depression questionnaire; they should not all be labeled nonrespondents.

Task A4 - Target population and inclusion/exclusion flow

Selection stage	Remaining N	Newly excluded
Matched DEMO-DPQ records	6,337	-
Age 20 years or above	6,064	273
Nine valid PHQ-9 items	5,210	854
Valid educational attainment	5,208	2
Complete age / sex / income / survey fields	4,584	624

Among the 6,064 adults, four records have missing or invalid education, but two of those had already been removed for incomplete PHQ-9. The main descriptive and unadjusted sample is 5,208; the preliminary complete-case sample for adjusted modeling is 4,584.

Selection criteria are sequential. Exclusion counts should not be summed independently across variables because missingness overlaps. The complete-case restriction may introduce selection bias, particularly if missing income is related to either education or screening status.

7. NHANES-specific requirements: survey design

Task A2 - Sampling weights and design features

WTINT2YR is the interview weight and WTMEC2YR is the MEC examination weight. Because the DPQ questionnaire is an MEC component, WTMEC2YR is used for descriptive weighting. SDMVSTRA and SDMVPSU encode masked sampling strata and primary sampling units for variance estimation.

15	30	0
Pseudo-strata	Stratum-PSU combinations	Missing/nonpositive MEC weights

WTMEC2YR among 5,208 participants	Value
Mean	40,585.81
Median	32,038.52
Minimum	5,424.59
Maximum	227,108.30

Sampling weights differ because selection probabilities, nonresponse adjustments, and population calibration vary. Point estimates can be weighted; valid standard errors, confidence intervals, and tests also need the complex design and appropriate survey methods.

Task A3 - Raw versus survey-weighted estimates

Screening estimate	Positive screening (%)
Unweighted sample proportion	12.67
MEC-weighted proportion	12.19
Weighted minus unweighted	-0.48 percentage points

The estimate changes because weighting gives different contribution to participants. This change is a difference between descriptive point estimates, not evidence of statistical significance. Educational subgroup comparisons are displayed in Figure 3 and the Task 9 table.

8. Final research question and interpretation boundaries

Task 10 - Final statistical question

Does the prevalence of positive depression screening (PHQ-9 total >= 10) differ by five-level educational attainment among U.S. adults aged 20 years and older, and does this association persist after adjustment for age, recorded sex, and the family income-to-poverty ratio?

The dataset contains the education exposure, PHQ-9 items, candidate adjustment variables, and survey design fields required for subsequent analyses. This question is associational rather than causal, because exposure was not randomized and measurement is cross-sectional.

Interpretive cautions

- Positive PHQ-9 screening describes self-reported symptom burden, not confirmed major depressive disorder.
- All hypothesis tests, adjusted effects, confidence intervals, and causal claims are outside Phase 1.
- The use of complete-case analysis can shift the analytical population; this requires sensitivity checks in later phases.
- The public income ratio is top-coded at 5 and age at 80; details in upper tails cannot be inferred.
- Appropriate NHANES survey variance estimation needs weights, masked strata, and PSUs; weights alone do not correct standard errors.

Planned analysis in Phases 2-5

Phase 2: Conditional probabilities and empirical/distributional probability models. Phase 3: Confidence intervals and hypothesis tests, with survey-design considerations. Phase 4: Adjusted regression assessing education while controlling for age, sex, and income. Phase 5: Integrated interpretation, sensitivity analysis, and final reproducible report.

Phase 1 conclusion

In the observed sample, higher education is generally associated with lower percentages of positive PHQ-9 screening. The absolute descriptive differences are substantial across the lowest and highest educational categories, but whether the association is statistically significant or remains after covariate adjustment is not answered by Phase 1.

Appendix. Deliverables and reproducibility

Phase 1 task coverage

Requirement	Where addressed
Tasks 1-3: provenance, unit, sampling	Section 1
Task 4: variable classification	Section 1
Tasks 5-6: quality and frequencies	Section 2
Task 7: center and dispersion	Section 3
Task 8: visualizations	Sections 3-5, Figures 1-4
Task 9: descriptive comparison	Section 5
Task 10: research question	Section 8
A1 and A4: merging and sample selection	Section 6
A2 and A3: weights and comparison	Section 7

Expected repository outputs

- Notebook: notebooks/phase1.ipynb (sequential, executable cells and narrative comments)
- Card: PROJECT_CARD.md; overview: README.md
- Processed data: data/processed/nhanes_adult_clean.csv (6,064), nhanes_phq9_analytical.csv (5,208), nhanes_complete_case.csv (4,584)
- Figures: figures/phase1/ (four PNG files)
- Report: reports/phase1.pdf; Git tag: phase-1

Reproduction and checks

Run the notebook from a fresh kernel after placing DEMO_L.xpt and DPQ_L.xpt under data/raw/. Confirm the three dataset sizes, uniqueness of SEQN, presence of figure outputs, and that all analytical code executes without errors. Keep report figures and numerical tables synchronized with the latest notebook run.

Primary documentation

NCHS / CDC. NHANES August 2021-August 2023 public files and cycle documentation.

DEMO_L - Demographic Variables and Sample Weights

DPQ_L - Depression Screener

NHANES Sample Design guidance

Source figures and summaries: user-provided Phase 1 notebook and processed data snapshot. The descriptive results are consistent with the latest participant counts shown in the notebook screenshot.